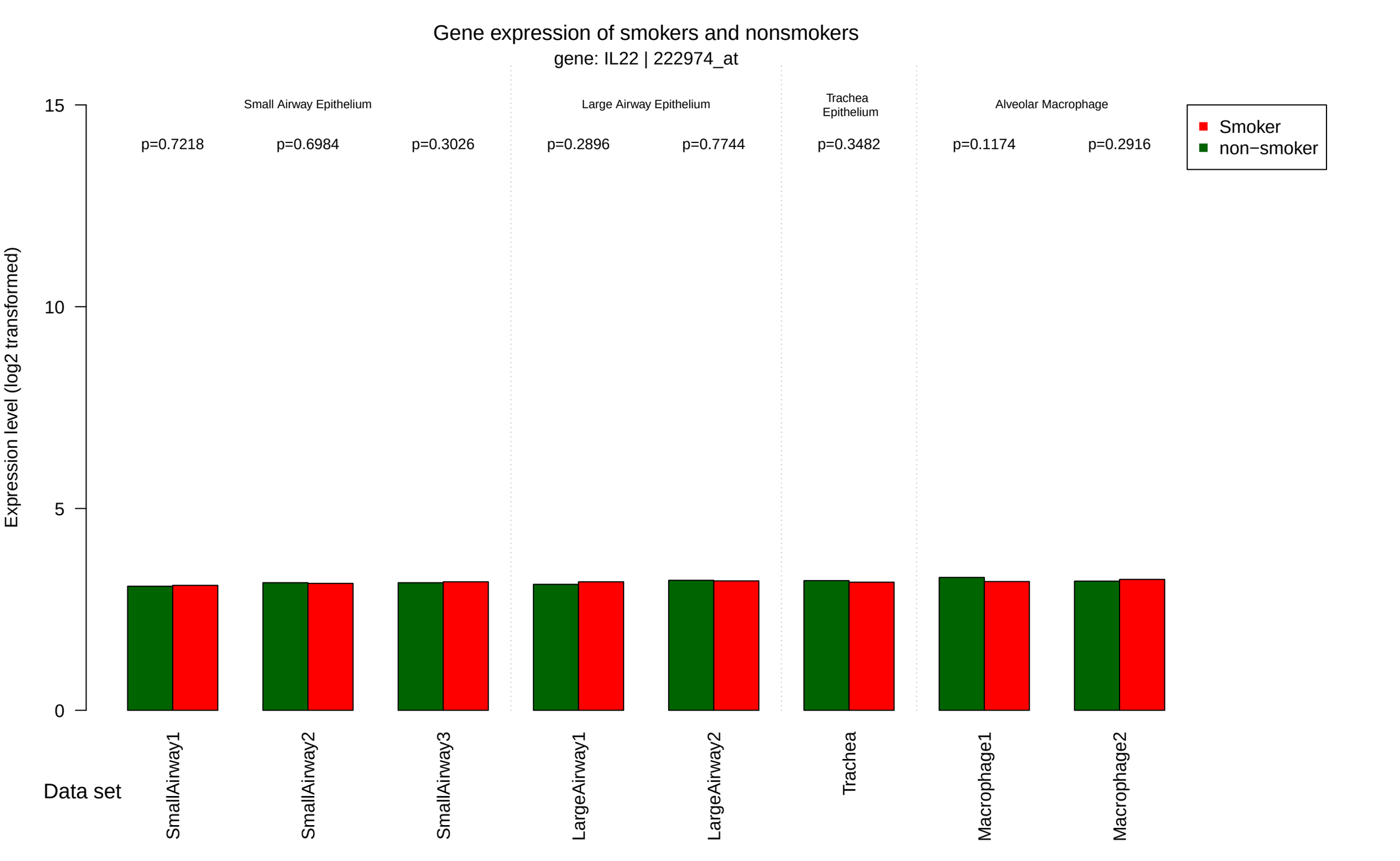
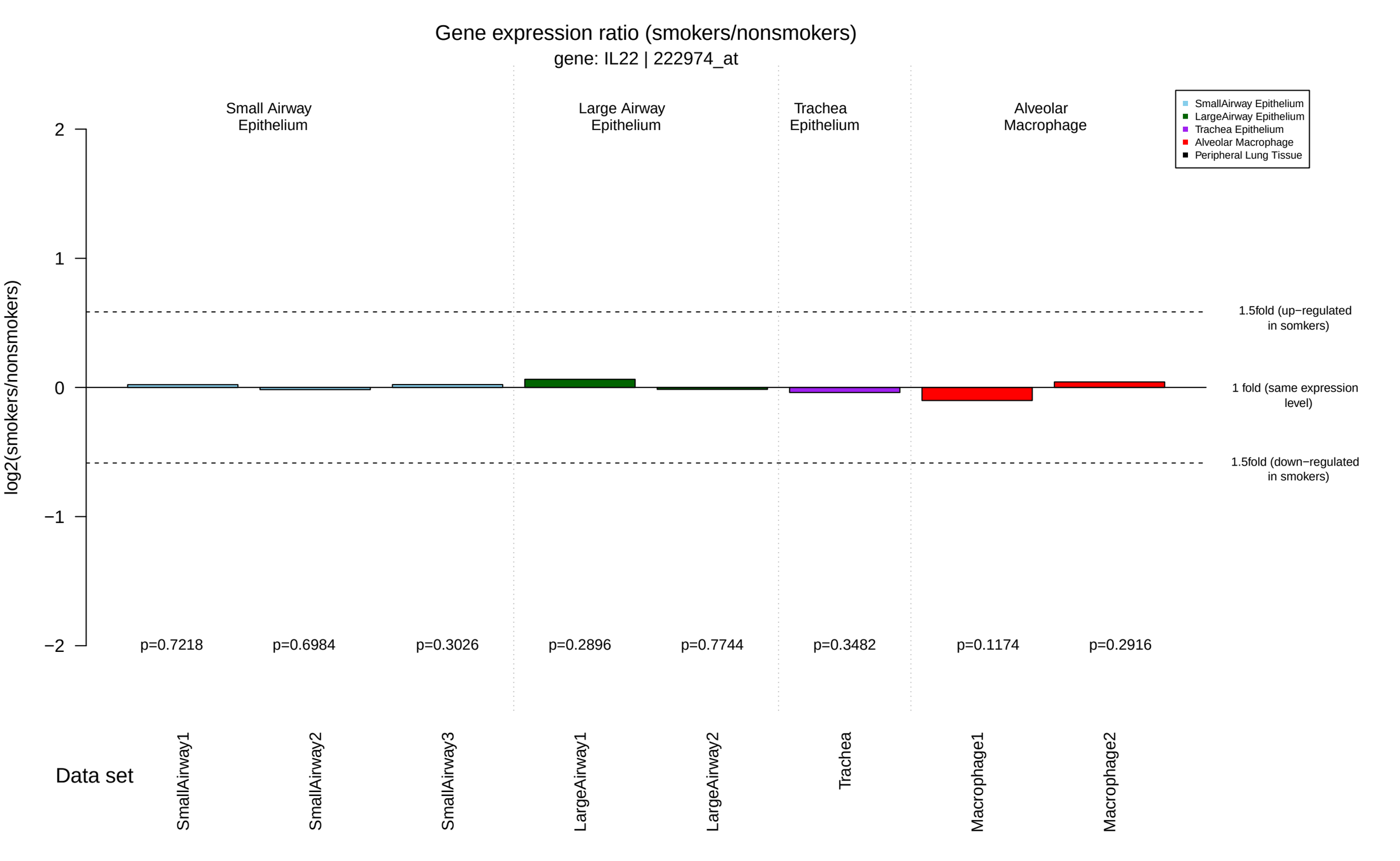
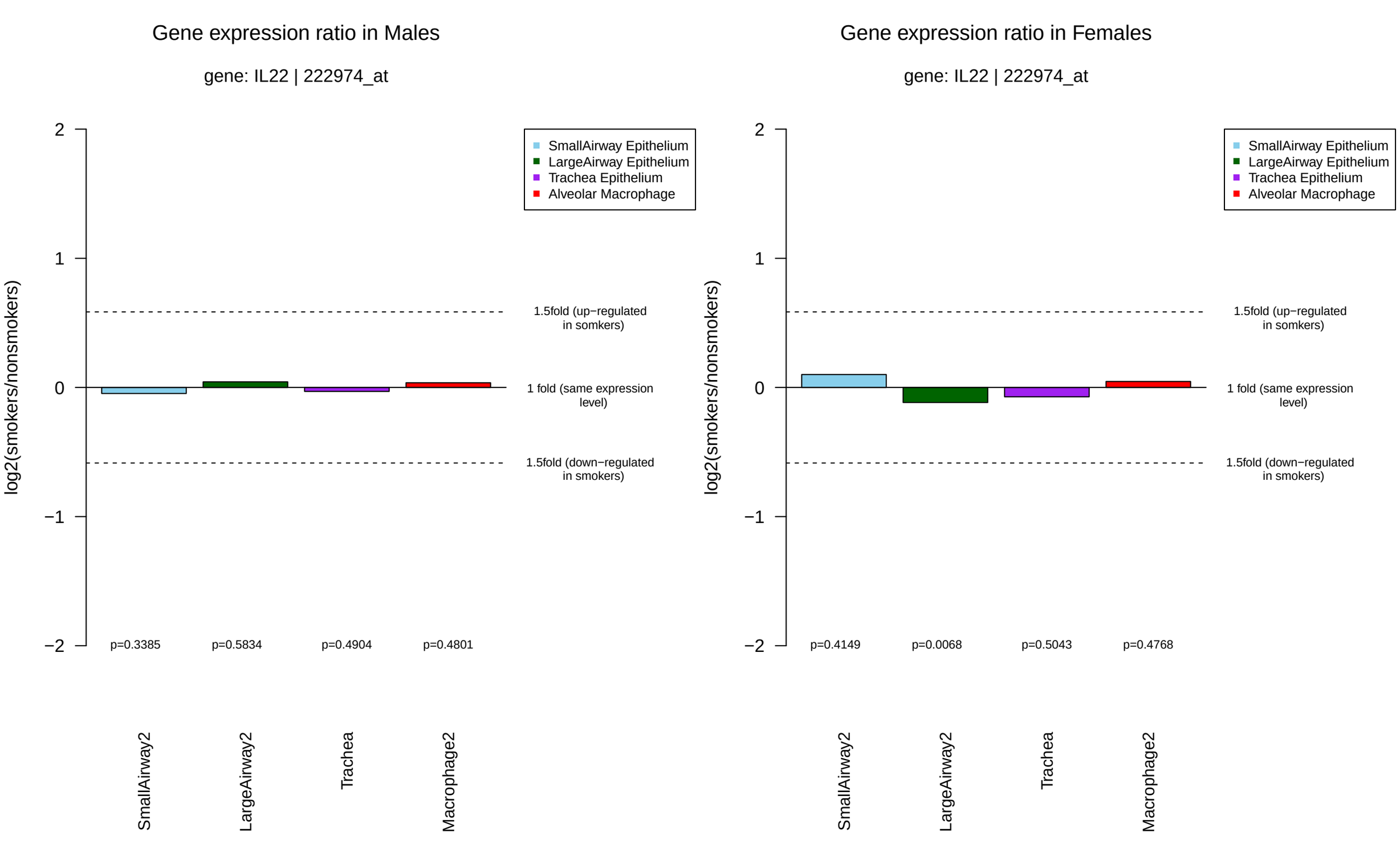
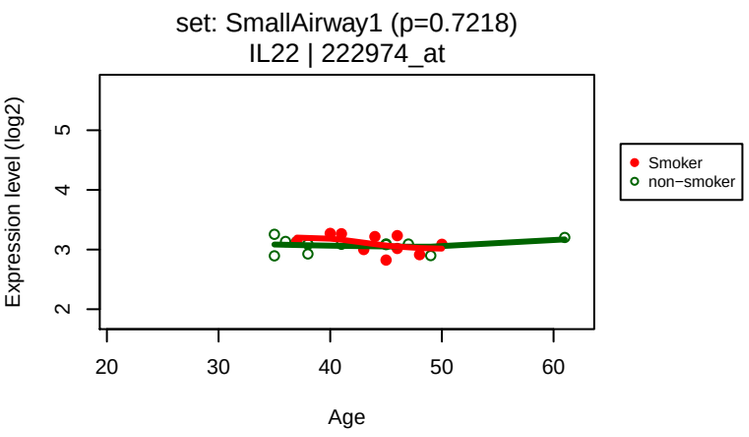
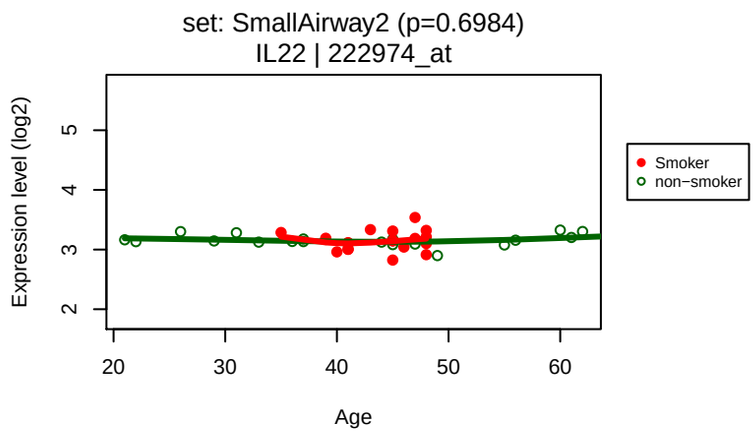
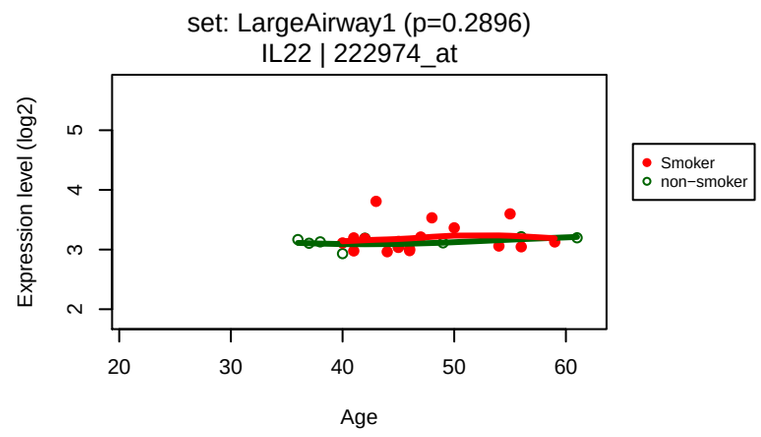

Gene expression of smokers and nonsmokers
gene: IL22 | 222974_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7218
p=0.6984
p=0.3026
p=0.2896
p=0.7744
p=0.3482
p=0.1174
p=0.2916
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

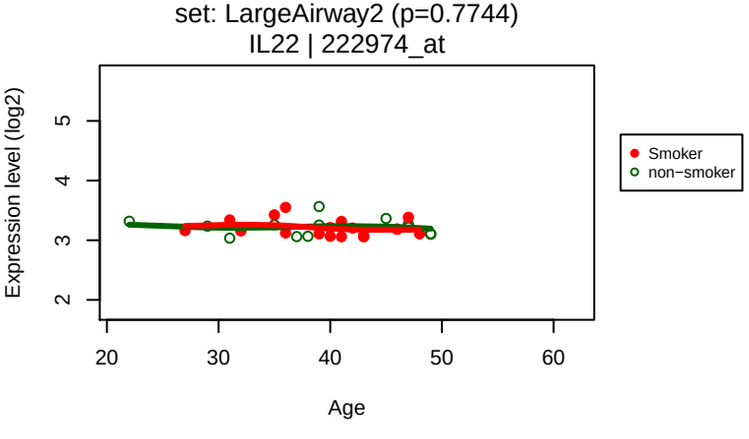

Gene expression ratio (smokers/nonsmokers)
gene: IL22 | 222974_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7218
p=0.6984
p=0.3026
p=0.2896
p=0.7744
p=0.3482
p=0.1174
p=0.2916
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

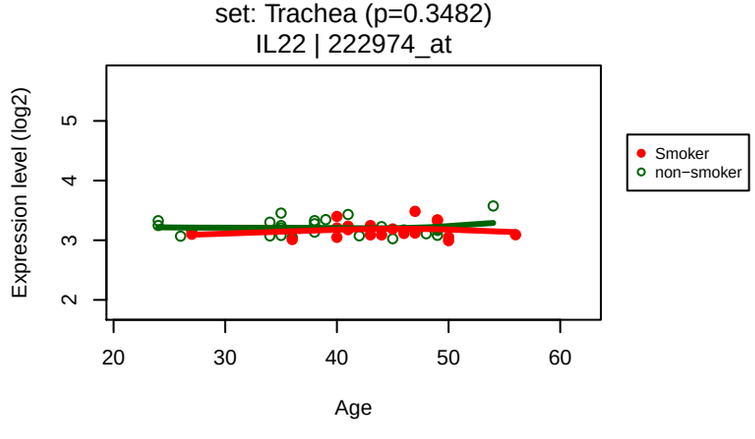

Gene expression ratio in Males
gene: IL22 | 222974_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3385
p=0.5834
p=0.4904
p=0.4801
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL22 | 222974_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4149
p=0.0068
p=0.5043
p=0.4768
SmallAirway2
LargeAirway2
Trachea
Macrophage2

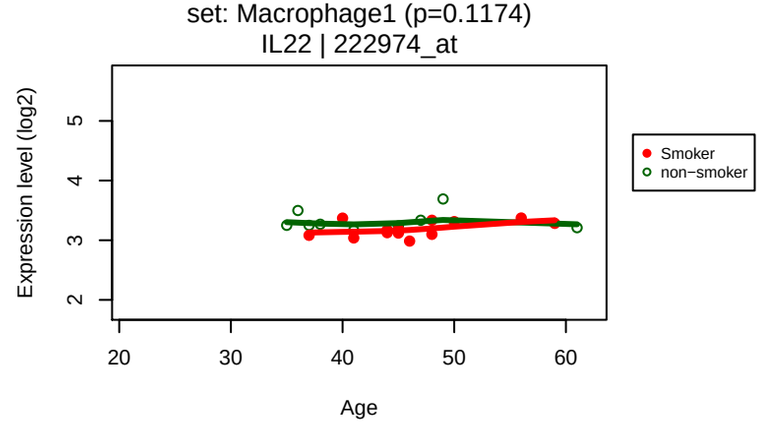

set: SmallAirway1 (p=0.7218)
IL22 | 222974_at
Expression level (log2)
Age
Smoker
non−smoker

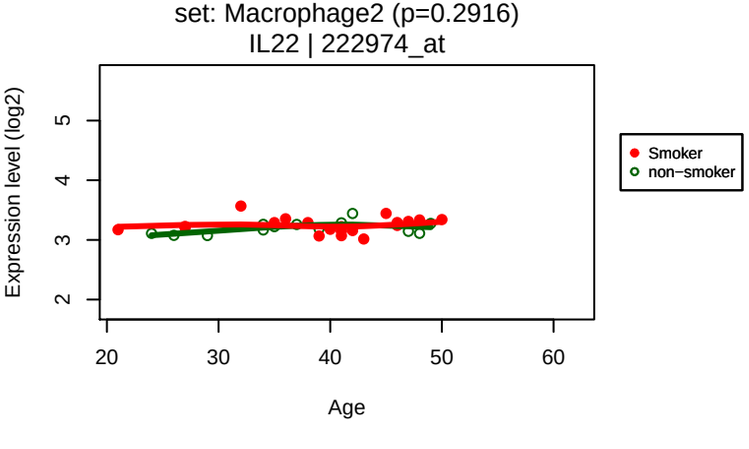

set: SmallAirway2 (p=0.6984)
IL22 | 222974_at
Expression level (log2)
Age
Smoker
non−smoker

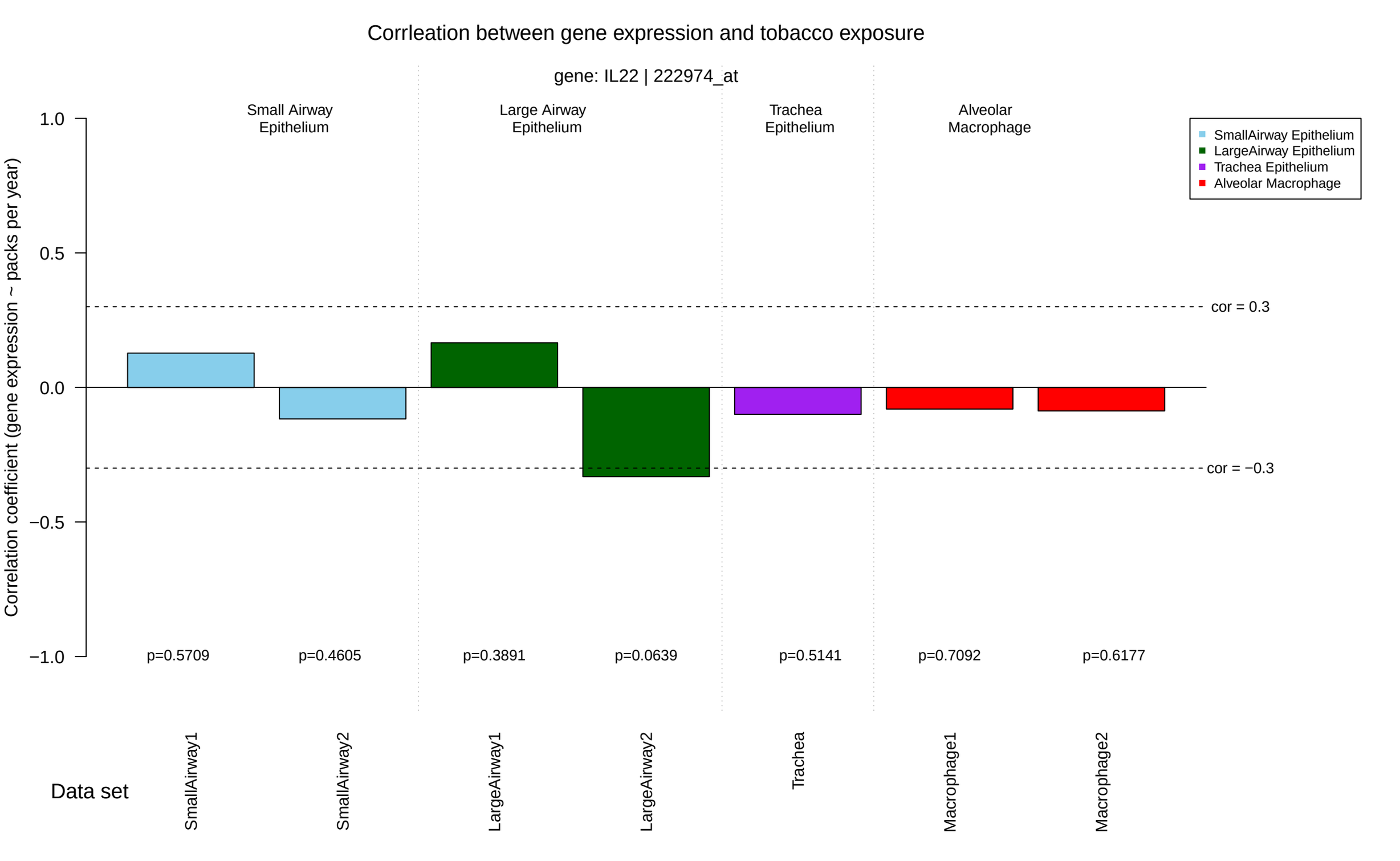

set: LargeAirway1 (p=0.2896)
IL22 | 222974_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7744)
IL22 | 222974_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3482)
IL22 | 222974_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1174)
IL22 | 222974_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2916)
IL22 | 222974_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL22 | 222974_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5709
p=0.4605
p=0.3891
p=0.0639
p=0.5141
p=0.7092
p=0.6177
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2